

NIYANTA AI — MASTER ARCHITECTURE & OPERATIONAL GUIDE

Platform: Niyanta AI Autonomous Traffic Intelligence & Security Platform | **Version:** v3.0.0 Enterprise
Document Type: Full Architecture, Operational Lifecycle, File Directory & Resilience Playbook

Welcome to **Niyanta AI**! This master guide provides a comprehensive manual covering the entire technical stack, component architecture, request lifecycle, complete file directory playbook, operational commands, and system resilience mechanisms.

Part 1: The Tech Stack & Security Suite Demystified

- 1. Docker & Docker Compose (Containerization):** Runs Python backend, Redis, and React frontend simultaneously.
- 2. Kubernetes (K8s) (Orchestration):** Deployment manifests for HPA scaling (3 → 50 pods) and kernel node-hardening.
- 3. FastAPI (The Web Framework & API Gateway):** High-performance async gateway handling all incoming traffic.
- 4. WAF Payload Inspector:** Intercepts URIs and POST body payloads to block SQL Injection (SQLi), Cross-Site Scripting (XSS), and Command Injection before routing.
- 5. Bot Fingerprinting (JA4):** Evaluates header cadences, User-Agent, and Sec-Ch-UA headers to isolate automated scripts (curl, python-requests, Puppeteer).
- 6. GraphQL Query Depth & Complexity Guard:** Restricts recursive GraphQL AST query nesting (max 7) and complexity scores (max 100) to block GraphQL DoS.
- 7. gRPC & HTTP/2 Binary Protocol Inspector:** Enforces multiplexed stream concurrency (max 100 per connection) and Protobuf boundary checks to prevent HTTP/2 Rapid Reset attacks.
- 8. Global Load Balancer (GLB) Multi-Region Router:** Computes live Anycast routing scores across global ingress regions (US East, EU West, AP South Mumbai).
- 9. Threat Intelligence Feeds:** Scans IP reputations against live AbuseIPDB, AlienVault OTX, and Tor Exit Node databases.
- 10. Chaos Engineering Studio:** Injects artificial fault modes (latency, packet loss, Redis outages) to verify automated auto-healing resiliency.
- 11. PPO Reinforcement Learning (PyTorch):** Dynamically computes allow, throttle, or block ratios based on host CPU, RAM, and latency.
- 12. Isolation Forest (The DDoS Detector):** Unsupervised anomaly detection flagging burst traffic spikes on Kafka event streams.
- 13. Downstream API Circuit Breaker:** Tracks downstream endpoint health (CLOSED, OPEN, HALF-OPEN) to shield databases during downstream failures.
- 14. Real-Time Webhook Alerts:** Automatically dispatches threat alerts to Slack, Discord, or Telegram channels.
- 15. SIEM Log Exporter:** Formats Common Event Format (CEF) and JSON security event logs for Splunk and Datadog.
- 16. Attack Replay Studio:** Allows security analysts to run 'What-If' policy simulations on attack snapshots directly from the UI.
- 17. ChromaDB & OpenAI (Explainability / RAG):** Generates human-readable audit logs in plain English.
- 18. React Dashboard:** Live telemetry UI featuring real-time charts, GLB routing status, Threat Intel feeds, WAF stats, bot ratios, and one-click SOC2 CSV/JSON report exports.

Part 2: The Complete Request Lifecycle

When a client makes a request to Niyanta AI, it flows through the security components in this exact order:

[Client Request]



1. WafScannerMiddleware ██████████ - Scans query/body for SQLi, XSS, and Command Injection.
 - Validates GraphQL query depth & complexity limits.
 - (Blocks with 403 / 400 if malicious or recursive)
2. ZeroTrustMiddleware ██████████ Validates JWT Bearer & Replay Timestamp signature.
3. NetworkProtectionMiddleware ████ - Checks Slow Loris, BCP38, RED, and CoDel delay headers.
 - Enforces gRPC / HTTP/2 multiplexed stream caps (100 max).
4. TrafficGatewayMiddleware ████████ - Evaluates Tiered API Key Quotas (Free, Pro, Enterprise).
 - Injects RFC 6585 headers (X-RateLimit-Limit, Remaining, Reset, Retry-After).
 - Checks Shadow Mode: if active, logs AI risk silently without blocking.
 - Queries Threat Intel feeds (AbuseIPDB / AlienVault).
 - Runs PPO Agent & Redis Token Bucket rate limiter.
5. Downstream Circuit Breaker ████ If endpoint healthy (CLOSED), forwards request to database.
If failing (OPEN), returns 503 fallback.
6. Asynchronous Logging ██████████ Emits telemetry to Kafka & SIEM Exporter (CEF/JSON).
7. React UI & Webhooks ██████████ Streams live updates to React Dashboard & dispatches Slack alerts.

Part 3: Folder & File Playbook

File Base Name	Relative Path	Purpose & Module Responsibilities
main.py	backend/app/main.py	Entrypoint. Configures FastAPI middlewares, router mounts, and startup handlers.
gateway.py	backend/app/middleware/gateway.py	Core Traffic Gateway (Token Bucket + RFC 6585 + Shadow Mode).
waf_scanner.py	backend/app/middleware/waf_scanner.py	WAF Payload Inspector (SQLi, XSS, Cmd Injection).
network_protection.py	backend/app/middleware/network_protection.py	RED, CoDel, Slow Loris, BCP38, H2 stream caps.
zero_trust.py	backend/app/middleware/zero_trust.py	Zero Trust HMAC/JWT and Request Replay validation.
bot_detector.py	backend/app/services/bot_detector.py	Bot Fingerprinting & JA4 header scanner.
graphql_analyzer.py	backend/app/services/graphql_analyzer.py	GraphQL Depth & Complexity Guard.
glb_router.py	backend/app/services/global_router.py	Multi-Region Global Load Balancer Router.
threat_intel.py	backend/app/services/threat_intel.py	AbuseIPDB & AlienVault Threat Feeds.
grpc_inspector.py	backend/app/services/grpc_inspector.py	gRPC & HTTP/2 Binary Protocol Inspector.
circuit_breaker.py	backend/app/services/circuit_breaker.py	Downstream Endpoint Circuit Breakers.
alert_service.py	backend/app/services/alert_service.py	Slack/Discord/Telegram Webhook alert dispatcher.

siem_exporter.py	backend/app/services/siem_exporter.py	CEF / JSON SIEM log exporter.
distributed_limiter.py	backend/app/services/distributed_limiter.py	Redis Lua Token Bucket.
ppo_agent.py	backend/app/ml/ppo_agent.py	PPO PyTorch model + fallback heuristics.
anomaly_detector.py	backend/app/ml/anomaly_detector.py	Isolation Forest unsupervised DDoS scorer.
routes.py	backend/app/api/routes.py	FastAPI routes (/ai/model-status, /chaos/inject, /glb/status, /graphql/analyze, /threat-intel/status).
Dashboard.jsx	frontend/src/pages/Dashboard.jsx	Real-time traffic, latency, GLB Router & Threat Intel status.
ChaosStudio.jsx	frontend/src/pages/ChaosStudio.jsx	Chaos Engineering fault injection & auto-healing studio.
AiInsights.jsx	frontend/src/pages/AiInsights.jsx	AI Reasoning Insights & High-Level Model Telemetry Hub.
Security.jsx	frontend/src/pages/Security.jsx	Live WAF status, Bot ratios, Circuit Breakers & CSV export.
AttackReplay.jsx	frontend/src/pages/AttackReplay.jsx	What-If Attack Replay Simulation Studio.
Policies.jsx	frontend/src/pages/Policies.jsx	Webhook alerts & Shadow Mode controls.
Analytics.jsx	frontend/src/pages/Analytics.jsx	SIEM Log Exporter & decision charts.
test_phase3_features.py	scratch/test_phase3_features.py	Comprehensive test script for Phase 3 features.

Part 4: Operational Command Playbook

1. Launch Backend API Gateway

```
cd backend
..\venv\Scripts\activate
python -m uvicorn app.main:app --port 8000
```

2. Launch React Control Dashboard

```
cd frontend
npm run dev
# Open http://localhost:5173 in browser
```

3. Run Automated Feature Verification Test Suite

```
..\venv\Scripts\activate
python scratch/test_phase3_features.py
```

Part 5: Resilience & Degradation Matrix

1. What if an attacker tries SQL Injection or XSS?

- WafScannerMiddleware intercepts the URI or body payload, matches regex signatures, and drops the request instantly with **403 Forbidden** before running database queries.

2. What if an attacker tries recursive GraphQL query depth attacks?

- **GraphQLAnalyzerService** computes AST depth. If depth > 7 or complexity > 100, drops the query immediately with **400 Bad Request**.

3. What if a botnet rotates IP addresses?

- **BotDetectorService** analyzes header order and missing browser headers (Sec-Ch-UA, Accept-Language), flagging automated scripts even if IP address changes.

4. What if a cloud region experiences high latency or outage?

- **GlobalLoadBalancer** detects latency degradation or high CPU (> 85%), dynamically re-routing ingress traffic to the next optimal Anycast node (e.g. AP South Mumbai).

5. What if a downstream database service crashes?

- **CircuitBreakerManager** trips from **CLOSED** to **OPEN**, immediately returning **503 Service Unavailable (Circuit Breaker Open)** to shield the database until health probes pass.